

Yerman J. Merel G.

Panamá City, Panamá | (+507) 6295 – 9441 | ymerel_02@outlook.com | [Portfolio](#) | [LinkedIn](#)

Mechanical Engineer with experience in industrial and infrastructure projects, specialized in project engineering support, technical documentation, mechanical systems validation, and compliance-driven execution. Proven ability to solve field-level operational issues, standardize procedures, and ensure traceability of engineering deliverables under demanding jobsite conditions.

Target Roles: Project Engineer | Mechanical Project Engineer | Field Engineer | Industrial Engineer

CORE COMPETENCIES

- Field Engineering Support & Technical Coordination
- Technical Documentation & SOP Development
- Commissioning Support & Shutdown Maintenance
- Structural Validation & Engineering Calculations
- Industrial Piping Design (P&IDs, Isometrics)
- Compliance, Traceability & Risk Mitigation
- Cross-Functional Coordination (Engineering, QC, Operations)
- CAD Design (AutoCAD, Inventor)
- ANSYS, MATLAB, Python
- SAP Business Suite, MS Office (Excel, Word, PowerPoint)

PROFESSIONAL EXPERIENCE

MONCAN

Co-Founder & Director of Systems Engineering

Panama City, Panamá
Aug 2025 – Present

- Designed the company's operational architecture using systems engineering principles, aligning engineering execution with business strategy.
- Developed digital workflows to improve traceability and control of project-related information from initial client contact to engineering delivery.
- Built cost-analysis and resource allocation models to support decision-making and improve operational efficiency.
- Structured scalable internal processes to reduce operational friction and improve execution consistency.

Herrenknecht Tunnelling Services Panama Corp.

Project Engineer

Panama City, Panamá
Jan 2025 – Jul 2025

- Developed and standardized a global Standard Operating Procedure (SOP) for critical mechanical maintenance operations, improving repeatability and ensuring consistent execution across international projects.
- Performed structural validation calculations and functional analysis of mechanical assemblies, ensuring safe operation under high-demand industrial conditions.
- Ensured regulatory compliance and documentary traceability, reducing contractual risk exposure and improving audit readiness.
- Supported engineering reporting and technical coordination between field operations and project stakeholders.

Projects: Hudson River Project (USA), Railway Tunnel Project.

China Railway Tunnel Group Co., LTD

Field Engineer of Tunnel Boring Machine and Superficial Facilities (*Intern*)

Panama City, Panamá

Aug 2024 – Jan 2025

- Designed corrective engineering solutions for mechanical and piping systems under unstable terrain conditions, eliminating severe incidents and improving operational continuity.
- Developed standardized technical documentation adopted as reference material by Quality Control and Engineering teams, reducing execution errors by 20%.
- Produced P&IDs and isometric drawings for high-pressure fluid systems, ensuring compliance with ASME B31 standards.
- Supported inspections, pressure testing activities, and mechanical assembly verification to ensure reliability and safety.

Project: Panama Metro Line 3 (Tunnel).

EDUCATION

TECHNOLOGICAL UNIVERSITY OF PANAMA

B.Sc. in Mechanical Engineering

Relevant Coursework: Finite Element Analysis (FEA), Computational Modeling, Numerical Methods, Control Systems.

Panama City, Panamá

Jan 2018 – Ago 2025

TRAINING & COURSES

- Design of Thermofluidic System** - *Technology University of Panama (Aug 2023)*
- Fluid Technology, Hydraulic and Lubrication** - *HERRENKNECHT AG (Jun 2025)*

TECHNICAL PROJECTS & RESEARCH

Resonant Wind Turbine Mathematical Model (Finalist – National Scientific Initiation Conference, 2023)

- Developed a mathematical model to analyze resonance behavior in wind turbine structures.
- Presented results at a national-level scientific engineering conference.

LANGUAGES

Spanish (Native) | English (C1 – Full Professional Proficiency) | German (A2)